through measuring the extent to which language models are able to reproduce copyrighted/licensed material in their training corpus with common sampling methods used in prior works (Carlini et al., 2021).

There are many situations in which training on, and generating, copyrighted material may be legally acceptable under doctrines like *fair use* (Lemley & Casey, 2020). But the degree of acceptability depends on the use case of the language model, as well as factors including the transformativeness of the content and the amount of content reproduced. We refer the reader to other works discussing the legal subject matter (Sobel, 2017; Lemley & Casey, 2020; Burk, 2019; Gillotte, 2019; Franceschelli & Musolesi, 2022). Fair use is not a panacea, though. In the case of ComicMix (Index, 2020), creators wrote a book based on Dr. Seuss's *Oh, the Places You'll Go!* titled *Oh, the Places You'll Boldly Go!* The book mimicked the style of Dr. Seuss, but replaced the text and imagery with a story in a Star Trek theme. This case was not covered by fair use because the book was a derivative product in a similar market that matched the original work too closely. As in the ComicMix case, machine learning models can memorize and output derivative content that is not necessarily protected by fair use. It is thus important to measure the memorization of copyrighted material in models to guide the investigative resources placed to address this risk.

Here, we introduce experiments to examine models' abilities to generate verbatim content. We emphasize that our numerical results do not (unconditionally) quantify the degree of legal risk. At a high level, given a model and a prompt (typically some initial portion of a copyrighted/licensed material), we measure the extent to which the model is able to reproduce the completion. This is a simple test of models' regurgitation capabilities. Specifically, we compiled prompts from three sources: (i) 1k randomly sampled books from the BooksCorpus (copyrighted), (ii) 20 books from the BooksCorpus that also appear in a list of bestsellers (copyrighted),[52] and (iii) 2k randomly sampled functions from the linux kernel source (GPL licensed).[53] For (i), we took varying numbers of tokens at the beginning of randomly sampled paragraphs as prompts. For (ii), we repeated the previous procedure—this time only considering the first paragraph of each bestseller. Lastly, for (iii), we took varying numbers of lines starting from the top of each function to form prompts. (i) and (ii) respectively test models' ability to reproduce the "average" content and content that is likely to be repeated in the training corpus. The comparison of the two is motivated by observations in past work that extraction of duplicated content is on average more likely (Kandpal et al., 2022).[54] Our evaluation metrics measure both exact regurgitation (longest common sequence normalized by length of given prefix) and near-exact reproduction (edit distance and edit similarity (Lee et al., 2021) normalized by length of given prefix).

Due to token credit constraints, we only sample one completion per prompt to test extraction. Thus, the results here may be underpowered as ideally one would generate many samples for the same prefix to approximate the worst-case behavior. In addition, given that we only focus on extracting selected contents (books and linux source code), our results may not fully reflect the qualitative behavior of extracting other sources. Nonetheless, we find that some models, particularly larger models and models trained on source code, generate verbatim content in certain situations. Future work with black-box model access can explore this further. We discuss additional nuances of this evaluation in the appendix.

5.5 Disinformation

Disinformation refers to false information that is disseminated by an actor with the intent to deceive, mislead, or otherwise influence the behavior of the target. Disinformation is of social concern: it has been used to disrupt democratic processes, undermine public health campaigns, and incite genocidal violence (Partnership, 2021; Zarocostas, 2020; Whitten-Woodring et al., 2020). Structurally, effective disinformation relies on both (i) convincing content and (ii) network behavior to propagate disinformation (e.g. via social media platforms) (Benkler et al., 2018).

To understand disinformation, we begin by considering approaches used for disinformation that pre-date modern language models. Disinformation actors generally have two approaches outlined in DiResta et al.

[52] https://www.theguardian.com/news/datablog/2012/aug/09/best-selling-books-all-time-fifty-shades-grey-compare

[53] GPL licenses generally require that the source code of derivative work also be distributed under the same license, compared to other licenses (e.g. MIT) which don't enforce such distribution.

[54] We expect portions of some of the bestsellers to appear repeatedly in the training corpus.

(2022). One option is to hire people in-house, which is good for operational security, but these employees may lack the cultural or linguistic knowledge required to make effective disinformation. This is particularly relevant when discussing geopolitical disinformation that may involve parties of very different cultural/linguistic backgrounds (e.g. different nations). The other option is to hire freelancers in the target country; freelance authors have the cultural context to make effective disinformation, but can more easily compromise the operation's security.

Given the growing generative capabilities of language models, malicious use for disinformation is a specific risk arising from the potential dual use of a language model. Specifically, several works have identified that language models can be a secure, efficient, and effective means for producing content for disinformation operations (Radford et al., 2019; Buchanan et al., 2021; Bommasani et al., 2021, §5.2). Relative to existing approaches, models can be created and stored in-house, matching the operational security of in-house operations, and can be trained on data from the foreign population, providing the effectiveness of remote operations (see Horvitz, 2022).

Beyond existing approaches that may be bottle-necked by the speed of human authorship, machine-generated text is highly scalable. Furthermore, when teamed with human editors who iteratively edit the model's generations, refine their inputs, and fine-tune the system, language models may generate outputs that perform better than either the human or the model can manage by themselves (Buchanan et al., 2021) Thus, the bottle-neck for using model generations for disinformation is reliability: if model generations need extensive human post-editing, the cost and risk of using them might be comparable to hiring humans to author text in the first place (Goldstein et al., Forthcoming).

Problem Setting. In foundational work on the relationship between language models and disinformation, Buchanan et al. (2021) introduce a taxonomy of six phenomena where language models may prove to be service. Of these six phenomena, we choose to focus on two: *narrative reiteration* and *narrative wedging*. We do so because the threats posed by these specifically align most strongly with the settings that concern disinformation researchers (Pennycook et al., 2021), and because they prompt models for short generations, so a larger number can be efficiently evaluated.

Narrative reiteration tests the ability of a language model to advance a specified narrative. We test this by adapting models to generate headlines that support a given thesis statement. In this sense, narrative reiteration reuses core capabilities related to the ability to paraphrase and the controllability of a language model's generation.

Narrative wedging, on the other hand, tests the ability of a language model to generate messages that divide people based on group identity (e.g. race or religion). Such wedging can be clearly divisive, fragmenting or polarizing a community by exerting pressure on individuals' sense of belonging to their social groups. We test this by adapting models to generate social media posts to amplify social divisions as well as to encourage a certain group to take a certain action. Further, divisive language may be either overtly hostile or only implicitly hostile (such as code words, microaggressions, stereotypes, and dogwhistles; Fogal et al., 2018; Quaranto, 2022). Hence, we distinguish between covert and overt hostility in the models' generations.

Datasets. Since we follow Buchanan et al. (2021), we begin by considering their dataset design for both reiteration and wedging. For reiteration, we choose to deviate from their work because they use only a single prompt in their evaluation. Instead, we use the Misinformation Reaction Frames dataset of Gabriel et al. (2022), which provides headlines and theses on COVID-19 and climate change. From this dataset, we manually cluster 114 headlines about COVID and 147 headlines about climate change from the Misinformation Reaction Frames dataset into 38 and 49 clusters respectively. For each cluster, we write a thesis statement that all of the headlines in that cluster support (e.g. "COVID-19 is a man-made disease."). For wedging, we use the same 11 prompts introduced by Buchanan et al. (2021). These prompts encourage certain voting behavior (vote Democratic, Republican, or not at all) targeting religious groups (Christians, Jews, and Muslims), as well encourage division (e.g. anti-Black racism).

5.6 Bias

In §4.7: METRICS-BIAS, we discuss social bias measurement in the context of realistic use cases, namely in the context of model generations. We believe such measurement is most directly useful for measuring the bias-related harms associated with language models. However, most of the literature on bias measurement in NLP, which can be traced back to Bolukbasi et al. (2016), instead focuses on biases that are more intrinsic and less directly grounded in extrinsically relevant use cases. As we note in §4.7: METRICS-BIAS, the predictive validity of these measurements (that is, the extent to which they predict biases in downstream behavior) has been questioned in several works (e.g. Goldfarb-Tarrant et al., 2021; Steed et al., 2022). As Bommasani et al. (2021, §5.1) describe, bias evaluation for foundation models can either target intrinsic properties of the model or extrinsic behavior of the model once adapted for a particular downstream use case. Here, we complement our existing evaluation with a finer-grained assessment of bias, noting that this also helps ensure coverage of biases that may be pertinent even for tasks that involve minimal/no generation.

Dataset Selection. Nadeem et al. (2020) and Nangia et al. (2020) each introduced minimal pair evaluations for assessing the biases of language models. To date, these two datasets have been the primary datasets for evaluating biases in language models, as they provide significant control due to the minimal pair design (which is the same design used for the linguistic evaluations that we describe in §5.1.2: LANGUAGE-MINIMAL-PAIRS). However, Blodgett et al. (2021) provides an extensive critique of these two datasets, providing comprehensive evidence for their inadequate validity.

For this reason, we instead turn to the more recently introduced **BBQ** dataset of Parrish et al. (2022). We note that the **BBQ** dataset may still suffer from some of the concerns discussed by Blodgett et al. (2021), but we expect it is comparatively better than the other options.[55] In particular, the **BBQ** dataset frames bias evaluation in the context of question answering, which also aligns with our more general approach of preferring evaluations that are more realistic and less synthetic when discussing social harms. **BBQ** measures biases associated with nine demographic categories, following the US Equal Employment Opportunities Commission, of age, disability status, gender, nationality, physical appearance, race/ethnicity, religion, socio-economic status, and sexual orientation. **BBQ** uses data generated via templates, with attested biases (with documented evidence that establishes the bias), that are then vetted by crowdworkers on Amazon Mechanical Turk.

Problem Setting. **BBQ** involves multiple-choice question answering, where each question is paired with a context and three answer choices (two of which reference different social groups of the same demographic category, and the third of which is always "Unknown"). To facilitate bias measurement, each example in the dataset is associated with a 2×2 pattern of instances: questions appear in pairs (one which is negative, i.e. a social value in the US is violated and the bias it reflects is harmful to certain groups, and the other of which is negative) and contexts appear in pairs (one which is ambiguous, and one which is disambiguating). Due to these alterations, in additional to measuring the exact match accuracy of the model on all data, the authors also introduces measures for the bias of the model both on ambiguous and disambiguated contexts.

5.7 Toxicity

In §4.8: METRICS-TOXICITY, we discuss toxicity measurement in the context of realistic use cases, namely in the context of model generations. In these use cases, we empirically find very little incidence of toxicity, which we take to be a generally positive sign regarding the risks of toxicity-related harms arising from language model deployment. On the other hand, Abid et al. (2021) demonstrate models have strong tendencies towards toxicity even with innocuous prompts, including in ways that reflect abhorrent social biases (e.g. associating Islam with violence). Further, it has been made clear by Kilcher (2022) that these models can be further intensified in their toxicity, which has created controversy in the AI research community over potentially reckless deployment (Liang & Reich, 2022). In light of this, we complement our existing evaluation with a finer-grained assessment of toxicity, noting that this also poses interesting connections with the models ability to detect toxicity (§3.7: TOXICITY-DETECTION).

[55] Parrish et al. (2022) do provide some discussion on this front.